TML Assignment 1 Report

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1 Introduction

This report presents our approach and findings for Assignment 1 of the TML course. Our primary objective was to evaluate and compare the performance of multiple membership inference attacks.

2 Introduction

There are 4 python files and 1 jupyter notebook that are important:

1. **simple_attack.py**: This file contains simple attack method implemented (see section **??**)
2. **main_lira**: This file contains LiRA method implemented (see section **??**)
3. **main_rmia**: This file contains RMIA method implemented (see section **??**)

3 Modeling Approach

To Evaluate which attack performs the best, we prepared 3 attacks:

3.1 Simple Attack:

In simple attack we divide the public dataset into train and test sets and use train ML models to decide members and non-members. In this approach we use features like entropy, loss, gradients, and confidence. To evaluate various machine learning models for attack prediction, we compared four approaches: MLP, XGBoost, CatBoost, and AdaBoost. The **CatBoost model achieved the highest ROC-AUC score of 0.6645**, outperforming the others and was therefore selected as the attacker model. Its success can be attributed to its **efficient handling of categorical features**, robust regularization strategies, and optimized tree structures. These characteristics enabled CatBoost to generalize well while avoiding overfitting, making it particularly suitable for this task.

XGBoost followed closely with a ROC-AUC of 0.6613, demonstrating strong generalization capabilities through its gradient boosting framework. However, unlike CatBoost, XGBoost requires more manual preprocessing for categorical variables, which may have slightly limited its performance. The **MLP**, serving as a baseline deep learning model, attained a ROC-AUC of 0.6474. While MLPs can capture non-linear patterns, they often underperform on tabular data unless extensively tuned or provided with large datasets.

Finally, **AdaBoost** scored the lowest with a ROC-AUC of 0.6440. As a simpler ensemble method, it may have struggled with complex data relationships and shown greater sensitivity to noise. Overall, the comparison highlights **CatBoost's superior performance**, driven by its ability to automatically handle categorical data and maintain strong generalization across samples. We visualize these results in the following bar chart:

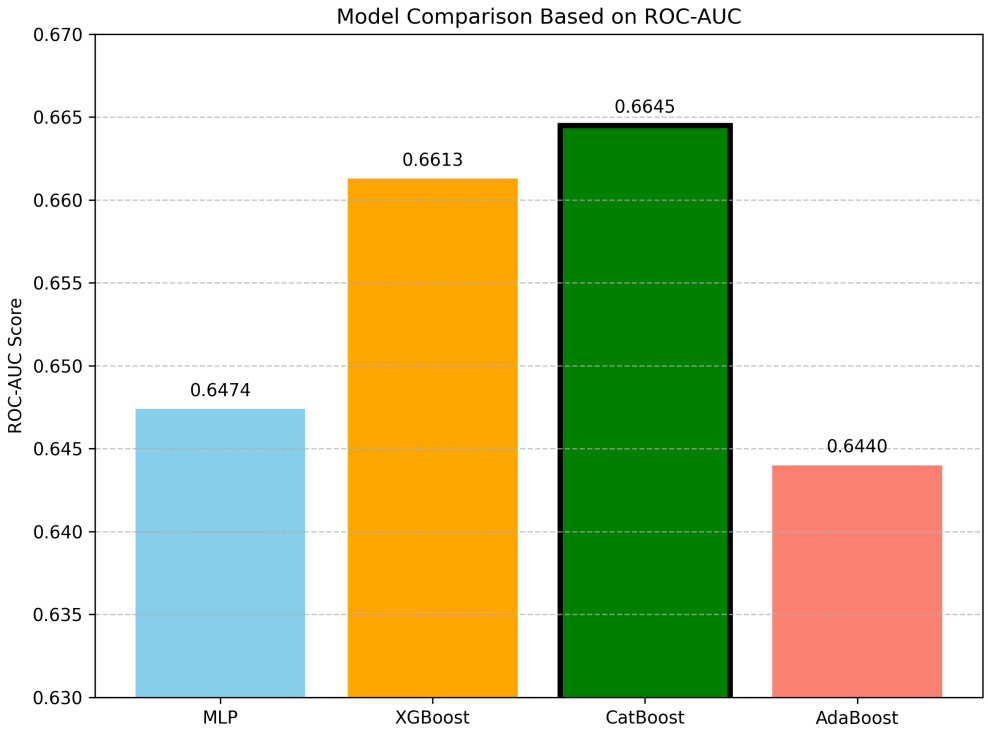

Figure 1: ROC-AUC comparison of evaluated models

3.2 LiRA

We used the LiRA algorithm to prepare another attack using one tailed hypothesis test (offline version of LiRA). Here we prepared 100-150 shadow models (the architecture used is the same as the target model). For each target model, we trained it on 0.8*len(public_dataset). We experimented by differing the value of number of shadow models. However, LiRA attacks gave us a **TPR@FPR=0.05 = 0.05 and an AUC of 0.54**

3.3 RMIA

We used the RMIA algorithm to prepare 3rd attack using offline version of RMIA. Here, too, we prepared 100-150 shadow models (the architecture used is the same as the target model). For the reference dataset we first took out 3000 samples from the public dataset. The length of this reference dataset is decided in accordance to the results mentioned in the appendix of the paper. We used gamma = 2 and a = 0.2. There values are optimal and have been mentioned int the paper. For each target model, we trained it on 0.8*len(remaining_public_dataset). We experimented by differing the value of number of shadow models. However, RMIA also gave us a **TPR@FPR=0.05 = 0.06 and an AUC of 0.52**

3.4 Combined Attack:

We tried combining all 3 attacks using RMIA and LiRA confidence values as additional features in the Simple Attack. However, this too gave worse results.

4 Conclusion

CatBoost achieved the highest ROC-AUC and was chosen as our primary attack model using simple features like loss, entropy and gradients. The performance improvement, though incremental, highlights CatBoost's

strength in handling complex feature interactions effectively.

Files in Repository

- `TML_Assignment_1.ipynb` - Main notebook
- `best_attack_model_catboost.pkl` - Saved CatBoost model
- `test.csv` - Provided test data
- `simple_attack.py`: This file contains simple attack method implemented
- `main_lira`: This file contains LiRA method implemented
- `main_rmia`: This file contains RMIA method implemented
- `roc_auc_comparison.png` - Performance chart
- `README.md` - Summary and guide

GitHub repository link: **Link**